Input-Anchored Cost Model for R/A/B/C Hamiltonian Application

Uni20 Developer Documentation

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1 Purpose

This note defines a restricted, implementation-oriented cost model for applying the two-site DMRG effective Hamiltonian in R/A/B/C form. It is a simplification of the more general hypergraph placement model in `docs/latex/rabc_hypergraph_partitioning.tex`. The aim is to remove most underdetermined communication choices while keeping the dominant Hamiltonian scheduling decisions.

The effective Hamiltonian is represented by sparse terms

$$t = (r(t), a(t), b(t), c(t), f_t), \qquad R_{r(t)} \mathrel{+}= f_t A_{a(t)} B_{b(t)} C_{c(t)}. \tag{1}$$

The Hamiltonian Lanczos path has one important invariant: the input center vector B and output center vector R have the same block space and must use the same persistent layout, so the next matvec can consume the output without a relayout.

2 Anchoring Assumption

Let $\mathcal{D}$ be the device set and $\mathcal{K}$ be the center-block set. Choose one canonical owner

$$h : \mathcal{K} \to \mathcal{D}. \tag{2}$$

For Hamiltonian application, B_k is initially resident on $h(k)$ and R_k must finally be resident on $h(k)$.

The input-anchored rule is:

$$\text{the first GEMM for term } t \text{ runs on } h(b(t)). \tag{3}$$

The output rule is:

$$\text{the second GEMM for term } t \text{ runs on } h(r(t)). \tag{4}$$

These two rules remove the independent choice of where to compute first-stage products, where to assemble temporaries, and where to accumulate final outputs. Given a layout h and a left/right path choice for each term, the copy and temporary schedule is induced.

This is not the fully general communication schedule. It excludes moving the first GEMM away from the input block, moving the second GEMM away from the final output block, tree reductions, temporary recomputation instead of copying, and arbitrary environment replication. These are deliberate restrictions. They turn an ill-determined communication problem into a concrete cost functional.

3 Optimization Variables

For each term choose only its contraction side:

$$s_t \in \{L, R\}. \tag{5}$$

Here $s_t = L$ means left-first,

$$A_a B_b \quad \text{then} \quad (A_a B_b) C_c, \tag{6}$$

and $s_t = R$ means right-first,

$$B_b C_c \quad \text{then} \quad A_a (B_b C_c). \tag{7}$$

The full restricted decision is therefore

$$(h, \{s_t\}_{t \in \mathcal{T}}). \tag{8}$$

4 Induced DAG

The induced DAG uses conservative exact temporaries. The scalar coefficient f_t is applied after the first-stage GEMM product has been formed, while accumulating that product into the local temporary. Before the resident matvec begins, setup has already materialized every environment block needed on the devices where the GEMM kernels below will run.

4.1 Left-First Terms

For each device, first form the distinct left-first GEMM products needed by terms whose input block lives on that device:

$$G_L(d) = \{(a(t), b(t)) : s_t = L,\ h(b(t)) = d\}. \tag{9}$$

For every $(a, b) \in G_L(d)$, compute once on device d:

$$X^d_{a,b} = A_a B_b. \tag{10}$$

The block B_b is already on $d = h(b)$. The environment block A_a is made available on d by the setup procedure before the resident matvec is timed.

Then, for every left-first term t with $(r, a, b, c) = (r(t), a(t), b(t), c(t))$, accumulate on $h(b)$:

$$Q^{L,h(b)}_{r,c} \mathrel{+}= f_t X^{h(b)}_{a,b}. \tag{11}$$

For each nonzero local partial $Q^{L,d}_{r,c}$, copy it to $h(r)$ if $d \neq h(r)$ and accumulate it into the assembled temporary

$$Q^L_{r,c} = \sum_{d \in \mathcal{D}} Q^{L,d}_{r,c} \quad \text{on } h(r). \tag{12}$$

Then execute the second GEMM on $h(r)$:

$$R_r \mathrel{+}= Q^L_{r,c} C_c \quad \text{on } h(r). \tag{13}$$

The environment block C_c has also been made available on $h(r)$ by setup.

4.2 Right-First Terms

For each device, first form the distinct right-first GEMM products needed by terms whose input block lives on that device:

$$G_R(d) = \{(b(t), c(t)) : s_t = R,\ h(b(t)) = d\}. \tag{14}$$

For every $(b, c) \in G_R(d)$, compute once on device d:

$$Y^d_{b,c} = B_b C_c. \tag{15}$$

The block B_b is already on $d = h(b)$. The environment block C_c is made available on d by the setup procedure before the resident matvec is timed.

Then, for every right-first term t with $(r, a, b, c) = (r(t), a(t), b(t), c(t))$, accumulate on $h(b)$:

$$Q^{R,h(b)}_{r,a} \mathrel{+}= f_t Y^{h(b)}_{b,c}. \tag{16}$$

For each nonzero local partial $Q^{R,d}_{r,a}$, copy it to $h(r)$ if $d \neq h(r)$ and accumulate it into the assembled temporary

$$Q^R_{r,a} = \sum_{d\in\mathcal{D}} Q^{R,d}_{r,a} \quad \text{on } h(r). \tag{17}$$

Then execute the second GEMM on $h(r)$:

$$R_r \mathrel{+}= A_a Q^R_{r,a} \quad \text{on } h(r). \tag{18}$$

The environment block A_a has also been made available on $h(r)$ by setup.

5 Cost Functional

The objective is predicted makespan:

$$\min_{h,s} \max_{d\in\mathcal{D}} C_d(h, s). \tag{19}$$

For each device d,

$$C_d = C^{\text{gemm}}_d + C^{\text{accum}}_d + C^{\text{temp}}_d + C^{\text{overhead}}_d + C^{\text{memory}}_d. \tag{20}$$

5.1 GEMM Cost

Let $\Phi(A_a, B_b)$ denote the GEMM flop count for $A_a B_b$ and similarly for other products. The first-stage cost is

$$\begin{aligned} C^{\text{first}}_d = \alpha_{L,1} &\sum_{(a,b)\in G_L(d)} \Phi(A_a, B_b) \\ + \alpha_{R,1} &\sum_{(b,c)\in G_R(d)} \Phi(B_b, C_c). \end{aligned} \tag{21}$$

The second-stage cost is grouped by assembled temporaries:

$$\begin{aligned} C^{\text{second}}_d = \alpha_{L,2} &\sum_{\substack{(r,c):Q^L_{r,c}\neq 0 \\ h(r)=d}} \Phi(Q^L_{r,c}, C_c) \\ + \alpha_{R,2} &\sum_{\substack{(r,a):Q^R_{r,a}\neq 0 \\ h(r)=d}} \Phi(A_a, Q^R_{r,a}). \end{aligned} \tag{22}$$

Then

$$C_d^{\text{gemm}} = C_d^{\text{first}} + C_d^{\text{second}}. \tag{23}$$

The coefficients α should be fitted, not assumed from peak FLOPS, because small block GEMMs include cuBLAS dispatch and launch overhead.

5.2 Temporary Accumulation Cost

After forming each first-stage product, the scheduler scales it by f_t and accumulates it into a local temporary. This is lower order than the GEMM, but it can matter for many small symmetry blocks. Let $\text{bytes}(X_{a,b}^d)$ and $\text{bytes}(Y_{b,c}^d)$ be the temporary product sizes. A simple bandwidth-like model is

$$\begin{aligned} C_d^{\text{accum}} = \delta_L & \sum_{\substack{t\in\mathcal{T}\\ s_t=L\\ h(b(t))=d}} \text{bytes}(X_{a(t),b(t)}^d) \\ &+ \delta_R \sum_{\substack{t\in\mathcal{T}\\ s_t=R\\ h(b(t))=d}} \text{bytes}(Y_{b(t),c(t)}^d). \end{aligned} \tag{24}$$

The associated kernel-launch or loop overhead is represented separately by the count features below.

5.3 Environment Setup

The first GEMM needs one environment side on $h(b)$; the second GEMM needs the other side on $h(r)$. In this input-anchored resident matvec model, environment placement is part of setup, not part of the Hamiltonian-apply cost functional. The setup step must materialize the required distinct environment blocks:

$$U_A^{L,1}(d) = \{a(t) : s_t = L,\ h(b(t)) = d\} \qquad \text{for } A_a B_b \text{ products,} \tag{25}$$

$$U_C^{L,2}(d) = \{c(t) : s_t = L,\ h(r(t)) = d\} \qquad \text{for } Q_{r,c}^L C_c \text{ products,} \tag{26}$$

$$U_C^{R,1}(d) = \{c(t) : s_t = R,\ h(b(t)) = d\} \qquad \text{for } B_b C_c \text{ products,} \tag{27}$$

$$U_A^{R,2}(d) = \{a(t) : s_t = R,\ h(r(t)) = d\} \qquad \text{for } A_a Q_{r,a}^R \text{ products.} \tag{28}$$

The setup benchmark may separately measure bytes and latency for this materialization, but those costs should not be included in C_d when fitting or optimizing the steady-state Lanczos matvec.

5.4 Temporary Migration

Left-first temporaries are keyed by (r, c) and initially accumulated on every device that owns at least one contributing B_b. Define

$$S_{r,c}^L = \{h(b(t)) : s_t = L,\ r(t) = r,\ c(t) = c\}, \tag{29}$$

$$S_{r,a}^R = \{h(b(t)) : s_t = R,\ r(t) = r,\ a(t) = a\}. \tag{30}$$

The temporary migration cost charged to destination device d is

$$\begin{aligned} C_d^{\text{temp}} = \beta_P & \sum_{\substack{(r,c):Q_{r,c}^L\neq 0\\ h(r)=d}} \sum_{\substack{s\in S_{r,c}^L\\ s\neq d}} \text{bytes}(Q_{r,c}^L) \\ &+ \beta_P \sum_{\substack{(r,a):Q_{r,a}^R\neq 0\\ h(r)=d}} \sum_{\substack{s\in S_{r,a}^R\\ s\neq d}} \text{bytes}(Q_{r,a}^R). \end{aligned} \tag{31}$$

This model assumes local accumulation before transfer. That is not a proof of optimality, but it avoids shipping one temporary per term and makes the communication cost quadratic-order in block dimensions rather than cubic-order GEMM work.

5.5 CUDA and Scheduler Overheads

For small blocks, count features are necessary. A minimal overhead model is

$$C_d^{\text{overhead}} = \gamma_1 N_d^{\text{first\_gemm}} + \gamma_A N_d^{\text{term\_accum}} + \gamma_2 N_d^{\text{second}} + \gamma_P N_d^{\text{temp\_copy}} + \gamma_Z N_d^{\text{zero}}. \tag{32}$$

Here $N_d^{\text{first\_gemm}} = |G_L(d)| + |G_R(d)|$ is the number of cached first-stage GEMM products on d, $N_d^{\text{term\_accum}}$ is the number of term contributions accumulated into local temporaries on d, N_d^{second} is the number of assembled second-stage temporaries multiplied on d, $N_d^{\text{temp\_copy}}$ counts temporary migrations into d, and N_d^{zero} counts output or temporary buffers that must be initialized. These coefficients should be learned from no-trace fixture benchmarks or low-overhead CUDA event timing.

5.6 Memory Pressure

Let $M_d(h, s)$ be the peak live bytes for the resident environment working set, local partial temporaries, assembled temporaries, and output buffers on d. The production model should enforce

$$M_d(h, s) \leq M_d^{\max}. \tag{33}$$

A prototype model may instead add

$$C_d^{\text{memory}} = \mu M_d(h, s), \tag{34}$$

but hard capacity constraints are the correct final form.

6 Relation to Cut-Based Policies

This model does not require a one-dimensional block ordering. The layout $h(k)$ is a graph placement of logical center blocks. After h is chosen, the storage layer may pack all blocks assigned to a device into a contiguous slab. Contiguity is therefore a storage consequence, not part of the mathematical domain.

A cut-based search can still be used as a baseline by restricting h to a chosen ordering of $\mathcal{K}$, but the ordering must be stated explicitly. The cost functional above applies equally to arbitrary placements and therefore is the better target for local search, graph partitioning, or offline integer programming.

7 Environment-Construction Variant

For environment construction, the input and output families need not share one layout. Use separate maps

$$h_B : \mathcal{B} \to \mathcal{D}, \qquad h_R : \mathcal{R} \to \mathcal{D}. \tag{35}$$

The same input-anchored rule can run first GEMMs on $h_B(b)$ and second GEMMs on $h_R(r)$. The Hamiltonian constraint $h_B = h_R$ should not be imposed in that case.